Scope Management Plan Template

Scope Management Plan

DOST STARBOOKS: Whiz Challenge

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INTRODUCTION

The STARBOOKS Whiz Challenge is a gamified, offline-first educational platform developed for the Department of Science and Technology – Science and Technology Information Institute (DOST-STII). The project involves the full design, development, testing, and deployment of an improved version of the existing DOST-STII Whiz Challenge application, addressing its identified limitations, most notably its single game format, lack of diverse learning strategies, and absence of an administrative management interface.

This Scope Management Plan was created to establish how the project's scope will be defined, managed, and controlled throughout its entire lifecycle. It documents the approach and processes Group NEXUS will follow to ensure that all work necessary to deliver the STARBOOKS Whiz Challenge is included in the project, and that no unnecessary work outside the agreed boundaries is performed. Managing scope with discipline is especially critical for this project given the team's time constraints as student developers, the fixed June 2026 deployment deadline, and the risk of scope creep identified in the Sprint 4 and Sprint 5 retrospectives.

The plan follows the five-step Project Scope Management process defined by PMBOK — Collect Requirements, Define Scope, Create WBS, Verify Scope, and Control Scope — applied within the Agile/Scrum methodology used to manage the project. Requirements were collected through stakeholder consultation with DOST-STII across multiple sprint reviews, analysis of the legacy application's shortcomings, expert judgment from the development team, and facilitated group workshops. These inputs were formalized into the project's Scope Statement, Work Breakdown Structure in OpenProject, and the WBS Dictionary documented in this plan. Any project communication, decision, or change that pertains to what work is or is not included in the STARBOOKS Whiz Challenge must reference and adhere to this Scope Management Plan.

Scope Management Approach

For the STARBOOKS Whiz Challenge project, scope management is the primary responsibility of the Project Manager. The scope for this project is defined by the Scope Statement, the Work Breakdown Structure (WBS) managed in OpenProject, and the WBS Dictionary included in this plan. The scope is measured and verified through sprint review meetings with DOST-STII at the conclusion of each sprint, deliverable acceptance checklists, and work performance measurements tracked in OpenProject (including work hours logged, task status, and budget consumption per phase).

Proposed scope changes may be initiated by the Project Manager, any team member, or DOST-STII stakeholders. All change requests must be submitted to the Project Manager, who will evaluate the requested change, perform an initial impact assessment, and convene a review meeting with the team and the Project Sponsor (DOST-STII) before escalating to formal acceptance. Upon approval, the Project Manager will update all project documents in OpenProject and communicate the scope change to all stakeholders. The Project Sponsor (DOST-STII) is responsible for the final acceptance of all project deliverables and the overall project scope, as evidenced by their sign-off during sprint reviews and at final project handover.

Roles and Responsibilities

The Project Manager, Project Sponsor, faculty adviser, and all team members play key roles in managing the scope of the STARBOOKS Whiz Challenge. The table below defines the roles and responsibilities for scope management throughout the project lifecycle.

Name	Role	Responsibilities
DOST-STII Representatives	Project Sponsor	Accept or reject scope change requests; evaluate alignment of deliverables with DOST-STII educational goals; provide formal acceptance of project deliverables at each sprint review and at project closure
Kelly Dumbrique	Project Manager / Product Owner	Measure and verify project scope against the WBS; facilitate and document scope change requests; perform impact assessments of proposed changes; organize and lead sprint review and change control meetings; update all project documents upon

		approval of scope changes; communicate scope decisions to all stakeholders
Mr. Jose Eugenio Quesada	Faculty Adviser	Review project scope for technical rigor and feasibility; validate scope change requests; provide guidance on scope decisions during weekly adviser meetings
Arcielle Marie Gercan	Backend Developer / Team Member	Perform work within the defined WBS scope; identify and communicate scope gaps or required changes to the Project Manager; participate in impact assessments and sprint reviews
Shandrae Lois Quianzon	Frontend Developer / Team Member	Perform work within the defined WBS scope; identify and communicate scope gaps or required changes to the Project Manager; participate in impact assessments and sprint reviews
Janice Maxene Salipande	Scrum Master / QA Lead / Team Member	Perform work within the defined WBS scope; maintain OpenProject tracking and WBS updates; identify and communicate scope gaps or required changes to the Project Manager; participate in impact assessments and sprint reviews

Table 1.1, *Scope Management Roles and Responsibilities*

Scope Definition

The scope for the STARBOOKS Whiz Challenge was defined through a comprehensive, multi-stage requirements collection and analysis process aligned with the Agile/Scrum methodology used to manage the project. The process began with a thorough analysis of the limitations of the existing DOST-STII Whiz Challenge application, which was documented and validated through direct consultation with DOST-STII technical staff and content specialists. Key deficiencies identified included limited replayability (single quiz format), absence of diverse learning strategies, insufficient player performance tracking, and lack of an administrative management interface.

From this analysis, the project team — in collaboration with DOST-STII — developed the foundational requirements by conducting structured interviews and stakeholder feedback sessions across multiple sprint reviews. Input was also gathered from the Faculty Adviser to ensure technical completeness and feasibility. Expert judgment from the team's frontend (Flutter/Dart) and backend (Laravel/MongoDB) developers was used to assess the feasibility of each requirement and translate stakeholder needs into concrete system features. Product analysis was applied to the existing STARBOOKS platform to identify integration boundaries and establish what the new system should and should not include.

The scope definition was formalized through the **Scope Statement** document (developed using the Project Management Foundations framework), which captures the project goals, boundaries, deliverables, success criteria, assumptions, and constraints. This document, along with the **Work Breakdown Structure** in OpenProject and the **WBS Dictionary** in this plan, constitutes the official scope baseline for the project.

PROJECT SCOPE STATEMENT

Product Scope Description

The STARBOOKS Whiz Challenge Version 1.0 is an improved, gamified educational platform developed for DOST-STII, designed to make Science and Mathematics learning more interactive, accessible, and engaging for students ranging from elementary to senior high school who attend DOST events and science fairs. The system is an offline-first application deployed on existing DOST kiosks via a LAN-based client-server architecture, requiring no internet connectivity for core gameplay.

The platform consists of four game modes (Whiz Challenge, Whiz Memory Match, Whiz Puzzle, and Whiz Battle), a badge and reward system, a player leaderboard, and a web-based administrative dashboard for DOST-STII staff. The administrative dashboard provides tools to manage players and admins, configure quiz content and difficulty settings, monitor reward claiming, and view game analytics.

Product Acceptance Criteria

The project will be considered complete and accepted when all of the following criteria are met:

- All four game modes operate correctly on DOST kiosk hardware with stable, bug-free gameplay and accurate offline scoring.
- The system functions fully offline for all core features.
- The administrative dashboard enables non-technical DOST staff to manage content, view analytics, and process reward claims without external assistance.
- The application runs without performance degradation on existing STARBOOKS kiosk hardware.
- Pilot testing with students yields positive feedback on usability, engagement, and educational value.

- The final deployment contains no critical bugs, crashes, or data integrity issues.
- DOST-STII formally reviews and accepts the final product as meeting their standards, goals, and STARBOOKS environment requirements.
- All development, testing, and deployment are completed, and the system is fully launched by June 2026.

Project Deliverables

- **Gamified Learning Application (Kiosk Version):** A fully functional web-based application with four game modes (Whiz Challenge, Whiz Memory Match, Whiz Puzzle, Whiz Battle), deployable on DOST kiosks via LAN.
- **Player Module:** Registration, login, profile management, progress tracking, badge collection, reward claiming, and leaderboard functionality for student players.
- **Admin Dashboard:** A web-based management interface for DOST-STII staff, covering player and admin account management, quiz content management, difficulty configuration, reward management, and analytics visualization with export capability.
- **Database Structure:** A MongoDB database schema storing all player records, game results, badge and reward data, quiz questions, and analytics data, structured for local deployment with optional cloud syncing.
- **Software Requirements Specification (SRS):** Complete functional and non-functional requirements documentation, including UML diagrams (DFDs, use case diagrams, activity diagrams, sequence diagrams, deployment diagram, and ERD).
- **System Design Document (SDD):** Architecture, database design, hardware/software design, and security controls documentation.
- **Testing and Documentation Package:** Complete user manuals, setup and deployment instructions, and test cases for system validation.
- **Deployment Package:** A deployment bundle containing all application components, configured for installation on DOST kiosk hardware.

Project Exclusions (Out of Scope)

- No integration with the main STARBOOKS digital library system.
- No mobile or tablet version; the system is limited to kiosk and PC browser deployment only.
- No topic-specific categories for Whiz Memory Match; only default STARBOOKS-related icons are used.
- Only jigsaw-type puzzles are supported in Whiz Puzzle; other puzzle formats such as word games are not included.
- No image or game asset customization through the admin interface; game assets and settings are fixed.

- Whiz Battle is limited to 1v1 public matchmaking via game code; private rooms and other multiplayer formats are not supported in Version 1.0.
- No player-specific analytics; the admin dashboard displays only overall trends and summary statistics across all users.

Project Constraints

- **Time:** The project must be fully completed and deployed by June 2026.
- **Technology:** All tools, platforms, and frameworks must be free or open-source and must run reliably on existing STARBOOKS kiosk hardware.
- **Connectivity:** The system must operate offline first; only Whiz Battle matchmaking and AI question generation require internet access.
- **Collaboration:** Development and review cycles must accommodate DOST-STII's office schedule and availability for sprint reviews and content validation.
- **Team Capacity:** The development team consists of students with concurrent academic responsibilities, which constrains total available working hours per sprint (10 hours per week per member).

Project Assumptions

- Existing STARBOOKS kiosks meet the minimum hardware requirements to run the new application without hardware upgrades.
- DOST-STII will continue to provide access to documents, technical guidance, educational content, and timely feedback throughout all project phases.
- Students and administrators are assumed to have basic digital literacy skills sufficient to navigate the application without extensive training.
- All free and open-source frameworks and tools used (Flutter, Laravel, MongoDB) will remain stable and supported through the project's launch date.

Work Breakdown Structure

In order to effectively manage the work required to complete the STARBOOKS Whiz Challenge project, the scope is subdivided into individual work packages tracked in OpenProject. The WBS follows a deliverable-oriented, hierarchical structure organized across five project phases, consistent with the 100% Rule — the WBS captures the total scope of the project with no work omitted or duplicated across branches. The WBS is structured as follows:

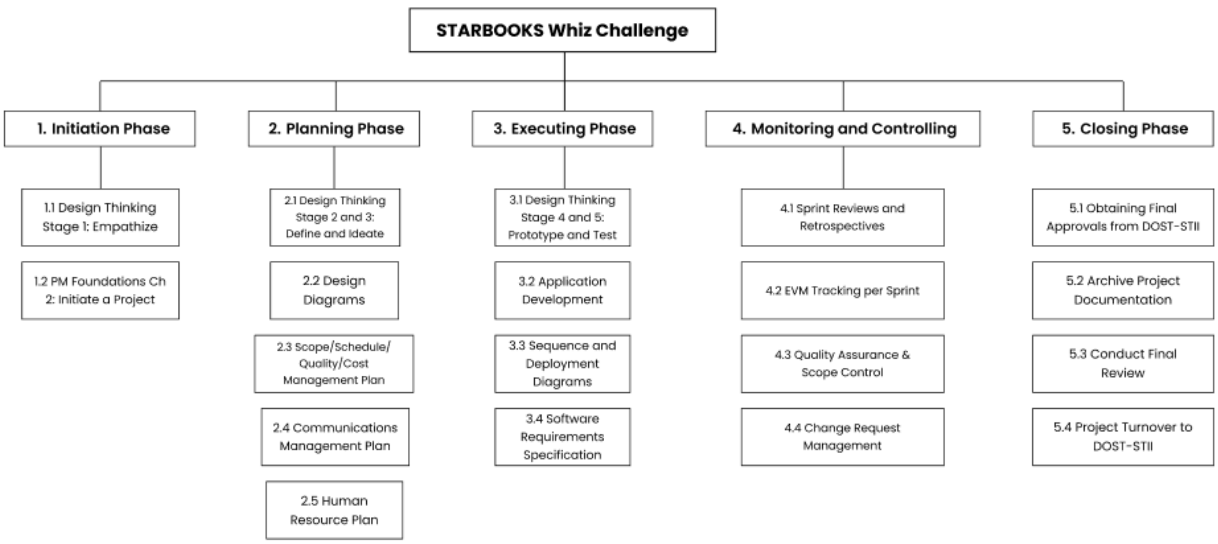

Figure 1.1, *Work Breakdown Structure (WBS)*

Each work package at the lowest level represents a discrete, assignable unit of work with a clear output, assigned owner, and tracked cost in OpenProject. The WBS Dictionary below provides a more detailed description, including deliverables, budget, and resources for each major WBS element.

Level	WBS Code	Element Name	Description of Work	Deliverables	Budget	Resources
1	1.0	Initiation Phase	All activities to formally start the project: client discovery, PM foundations initiation tasks, charter creation, objectives, scope,	Project Charter, Objectives Document, Scope Statement, Stakeholder Analysis, Initial	PHP 201,000	All 4 team members

			stakeholder analysis, budget and timeline definition	Budget and Timeline		
2	1.1	Design Thinking: Empathize	Conduct empathy research with DOST-STII to understand user needs, challenges with the legacy system, and target audience characteristics	Empathy research output, user personas	PHP 10,200	Janice Maxene Salipande
2	1.2	PM Foundations: Initiate	Complete PM Foundations Ch 2 deliverables including Charter, Objectives, Scope, and Stakeholder Analysis	Completed PM Foundations documents	PHP 1,000	All 4 team members
2	1.3	Define Budgets	Establish phase-level budget allocations and document them in OpenProject	Phase budget documentation in OpenProject	PHP 1,000	Janice Maxene Salipande
1	2.0	Planning Phase	All activities to produce project planning artifacts: design thinking, system diagrams (DFDs, use cases, ERD, deployment), and knowledge area management plans	All planning documents, system diagrams, management plans	PHP 201,800	All 4 team members
2	2.1	Design Thinking: Define	Synthesize empathy findings, define the problem	Problem Statement, Design Brief	PHP 2,500	Arcielle Marie Gercan

			statement and project goals			
2	2.2	Design Thinking: Ideate	Brainstorm and evaluate design solutions for the application	Ideation outputs, selected design direction	PHP 2,500	Shandrae Lois Quianzon
2	2.3	Design Diagrams	Produce all system analysis and design diagrams: DFDs (Context, Level 1, Level 2), Use Case Diagrams with fully dressed use cases, activity diagrams, sequence diagrams, test cases, ERD, and deployment diagram	Complete SRS-ready diagram set	PHP 62,500	All 4 team members
2	2.4–2.13	Management Plans	Produce all subsidiary management plans (Scope, Schedule, Quality, Resource, Communications, Risk, Procurement, Stakeholder, Change)	Completed management plan documents	TBD	All 4 team members
1	3.0	Executing Phase	Full-scale development and testing of the STARBOOKS Whiz Challenge application, covering all player-side features, all admin dashboard features, and	Fully functional application, SRS, Deployment Diagram, Sequence Diagrams	PHP 1,002,600	All 4 team members

			supporting documentation			
2	3.3.1	Registration Page	Build multi-step player registration with form validation, API integration, password hashing, and database storage	Completed Registration Page feature	PHP 60,000	Arcielle Marie Gercan
2	3.3.2	Player Login Page	Build player login with credential validation, session/token management, and error messaging	Completed Player Login Page feature	PHP 14,000	Arcielle Marie Gercan
2	3.3.3	Homepage with Player Profile	Build homepage with player profile section, badge display, avatar, and navigation to four game modes	Completed Homepage feature	PHP 70,000	Shandrae Lois Quianzon
2	3.3.4	Player Leaderboard	Build leaderboard view with rankings, avatar display, stats breakdown, and toggle between game modes	Completed Player Leaderboard feature	PHP 120,000	Shandrae Lois Quianzon
2	3.3.5.1	Whiz Challenge	Build solo quiz game mode with category/difficulty selection, randomized questions, timer, scoring, and badge award logic	Completed Whiz Challenge game feature	PHP 60,000	Shandrae Lois Quianzon

2	3.3.5.2	Whiz Memory Match	Build card-matching game with randomized grids, flip/match logic, timer, move counter, and star scoring	Completed Whiz Memory Match game feature	PHP 42,000	Arcielle Marie Gercan
2	3.3.5.3	Whiz Puzzle	Build jigsaw puzzle game with image selection by category/difficulty, drag-and-drop mechanics, piece snapping, timer, and completion detection	Completed Whiz Puzzle game feature	PHP 21,000	Janice Maxene Salipande
2	3.3.5.4	Whiz Battle	Build real-time 1v1 quiz battle with game code generation, WebSocket-based player synchronization, per-question score updates, and badge award for winner	Completed Whiz Battle game feature	PHP 76,000	Kelly Dumbrique
2	3.3.6	Admin Dashboard	Build complete administrative dashboard including login, analytics, leaderboard, player management, admin management, question management, and difficulty settings	Completed Admin Dashboard feature set	PHP 260,000	All 4 team members

1	4.0	Monitoring & Controlling	Ongoing tracking of project progress, cost variance, schedule variance, quality checks, and sprint retrospectives throughout the project lifecycle	Sprint status reports, EVM metrics per sprint	PHP 202,400	All 4 team members
1	5.0	Closing Phase	All activities to formally close the project: completing outstanding tasks, obtaining final DOST-STII approvals, archiving documentation, conducting final review, and project turnover	Final accepted deliverables, archived project documentation, turnover package	PHP 202,200	All 4 team members

Table 1.2, *WBS Dictionary*

Scope Verification

As this project progresses through each sprint, the Project Manager will verify completed deliverables against the scope baseline defined in the Scope Statement, WBS, and WBS Dictionary. Verification occurs at two levels: sprint-level and phase-level.

At the sprint level, the Project Manager reviews completed work packages in OpenProject at the end of each two-week sprint, confirming that each task and feature meets its defined acceptance criteria before it is presented at the Sprint Review. The Sprint Review meeting with DOST-STII serves as the formal verification checkpoint for that sprint's deliverables. Feedback from DOST-STII during this meeting is documented and used to inform the next sprint backlog.

At the phase level, upon completion of each project phase (Initiation, Planning, Executing, Monitoring & Controlling, Closing), the Project Manager and Project Sponsor will meet to formally review and accept all phase deliverables. Formal acceptance is documented through a signed deliverable acceptance record, confirming that the completed work is consistent with the approved scope baseline. This incremental acceptance approach of verifying deliverables at each sprint and phase boundary rather than deferring acceptance to the end of the project, ensures that scope alignment is maintained continuously and that deviations are caught and addressed before they compound.

Scope Control

The Project Manager and the project team will collaborate to control the scope of the STARBOOKS Whiz Challenge throughout the project lifecycle. The WBS Dictionary serves as the primary statement of work for each WBS element, and team members are expected to perform only the work defined within each element and produce the stated deliverables. The Project Manager monitors project progress through OpenProject to ensure that this scope discipline is upheld.

Scope Change Process:
Any project team member, faculty adviser, or DOST-STII stakeholder may identify and propose a scope change. The following process governs all scope change requests:

1. The requestor documents the proposed change in a **Project Change Request**, describing the change, the reason for it, and the expected impact on scope, schedule, and budget.
2. The **Project Manager** reviews the change request and performs an initial evaluation to determine whether it falls within the intent of the project. Clearly out-of-scope requests are denied at this stage with documented justification.
3. If the change request passes initial review, the Project Manager convenes a **change control meeting** with the project team, faculty adviser, and DOST-STII to conduct a formal impact assessment covering scope, schedule, cost, and risk implications.
4. Upon initial approval by the Project Manager and DOST-STII, the Project Manager formally submits the request to the **Change Control Board** (comprising the faculty adviser and relevant DOST-STII representatives) for final review.
5. If the **Change Control Board** approves the scope change, the **Project Sponsor (DOST-STII)** formally accepts it by signing the Project Change Control Document.
6. Upon final acceptance, the **Project Manager** updates all relevant project documents (WBS in OpenProject, Scope Statement, management plans as applicable) and communicates the approved change to all team members and stakeholders.

Sponsor Acceptance

Approved by the Project Sponsor:

____ Date: April 8, 2026

Marievic V. Narquita
Information Services Section (ISS) Head of DOST-STII